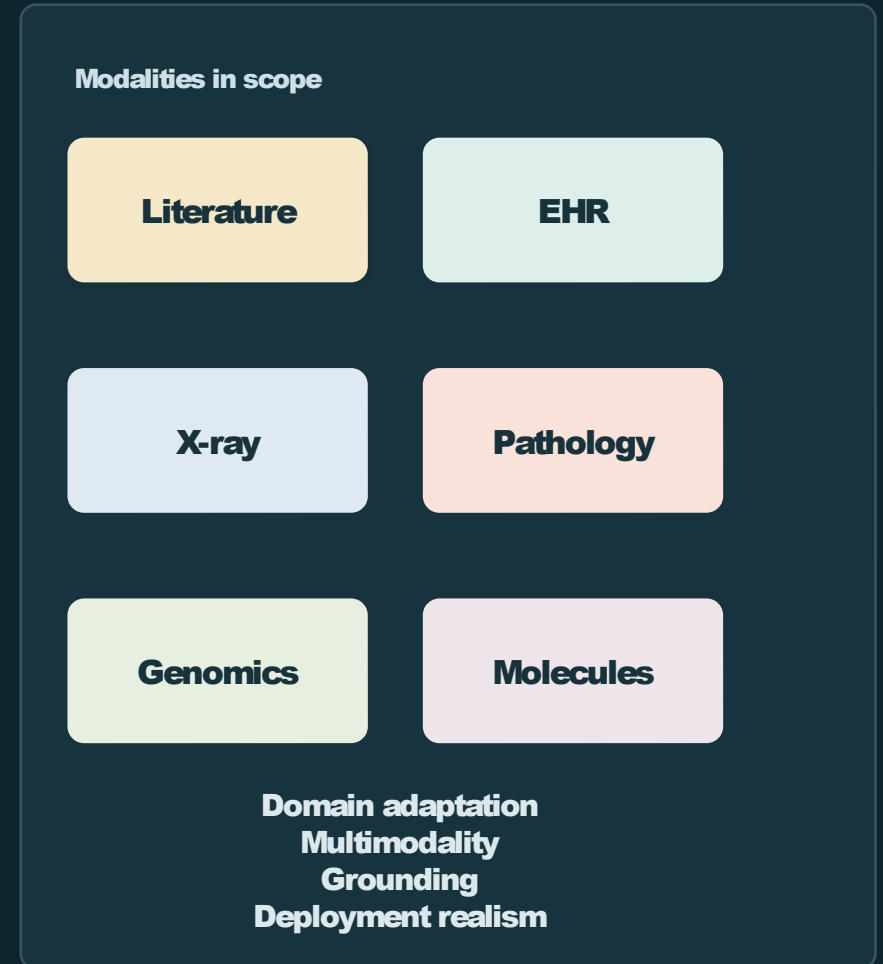


Biomedical LLMs

From PubMed to Multimodal Copilots

Theme: domain adaptation, multimodality, grounding, deployment realism



Why Biomedical LLMs Are Different

Language shift

Specialized jargon, abbreviations, and ambiguous synonyms make web-trained assumptions brittle.

Modality shift

Real biomedical reasoning mixes text with images, slides, priors, signals, sequences, and measurements.

Risk shift

A plausible answer can still be clinically unsafe. Evaluation, traceability, and workflow fit matter more.

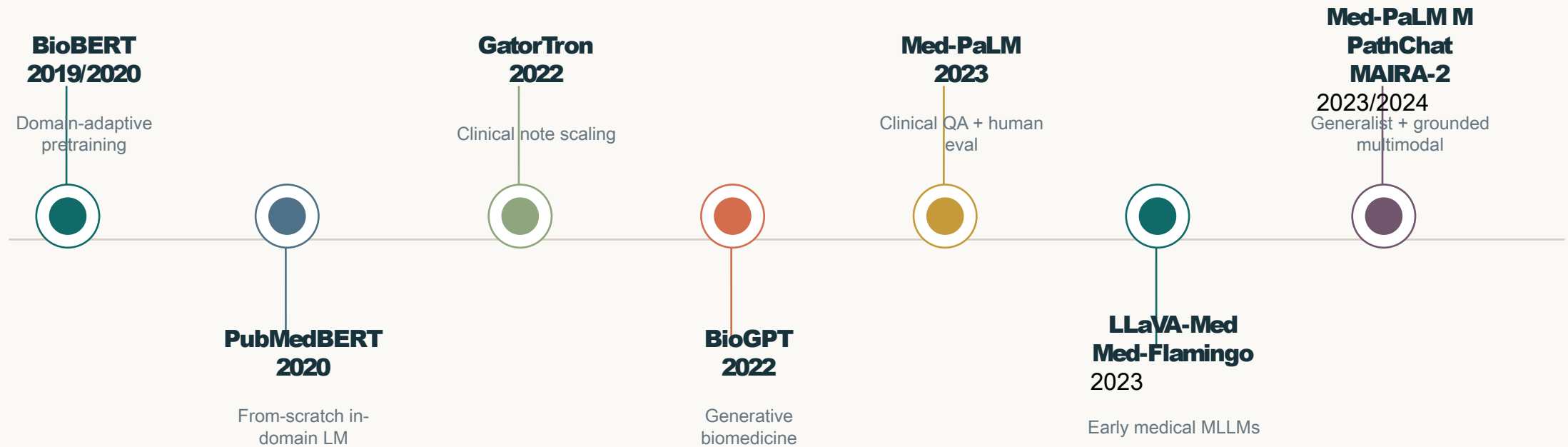
Teaching lens

Domain adaptation matters

Multimodality is native to medicine

Evaluation beats fluency headlines

Evolution Timeline



The storyline: domain-specialized text models → generative medical reasoning → multimodal, grounded assistants

Domain Adaptation Lesson: BioBERT and PubMedBERT

BioBERT

Continue pretraining from a general BERT on PubMed and PMC.

Teaching point: domain-adaptive pretraining works.



PubMedBERT

Train from scratch on biomedical text.

Teaching point: when the domain is distinct enough, in-domain initialization can win.

[Figure — see notes]

Pretraining pipeline and task transfer view

Source citation: Lee et al. 2020, BioBERT Figure 1; pair with Gu et al. 2020 for the from-scratch contrast.

Scaling Clinical Text: GatorTron

- Clinical notes are compressed, institution-specific, and privacy sensitive.
- Scaling helps, but strong clinical corpora are much harder to build than PubMed corpora.
- Clinical deployment questions appear earlier because the source text is closer to patient care.

Discussion prompt

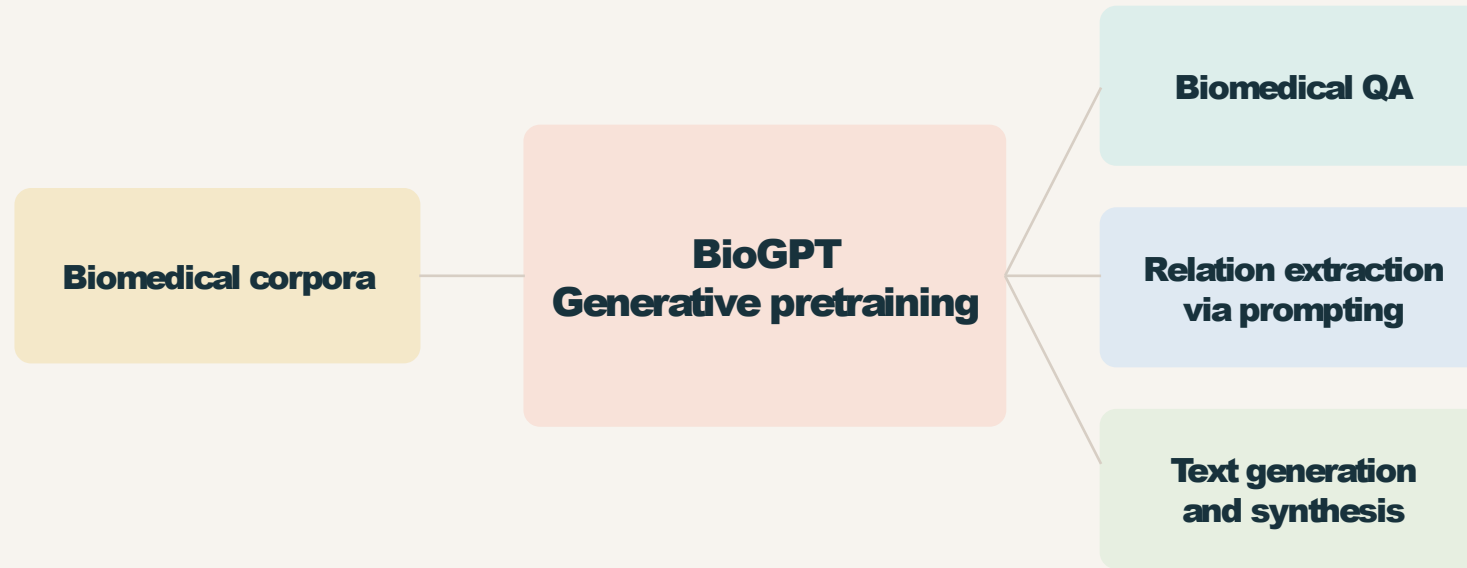
Why does scaling a clinical model raise different reproducibility questions than scaling a literature model?

[Figure — see notes]

Figure: GatorTron scaling vs. performance

Source citation: Yang et al. 2022, GatorTron Figure 3.

From Representation Models to Generative Biomedical LLMs



What changed?

The target shifted from features for downstream classifiers to open-ended generation. That increased flexibility, but it also increased the space for plausible hallucinations.

Bridge statement

This is the transition from “language model as encoder” to “language model as biomedical assistant.”

Medical QA and Alignment

Med-PaLM, Med-PaLM 2, and MEDITRON

Med-PaLM

Benchmark plus human evaluation.

Med-PaLM 2

Stronger medical QA and expert comparison.

MEDITRON

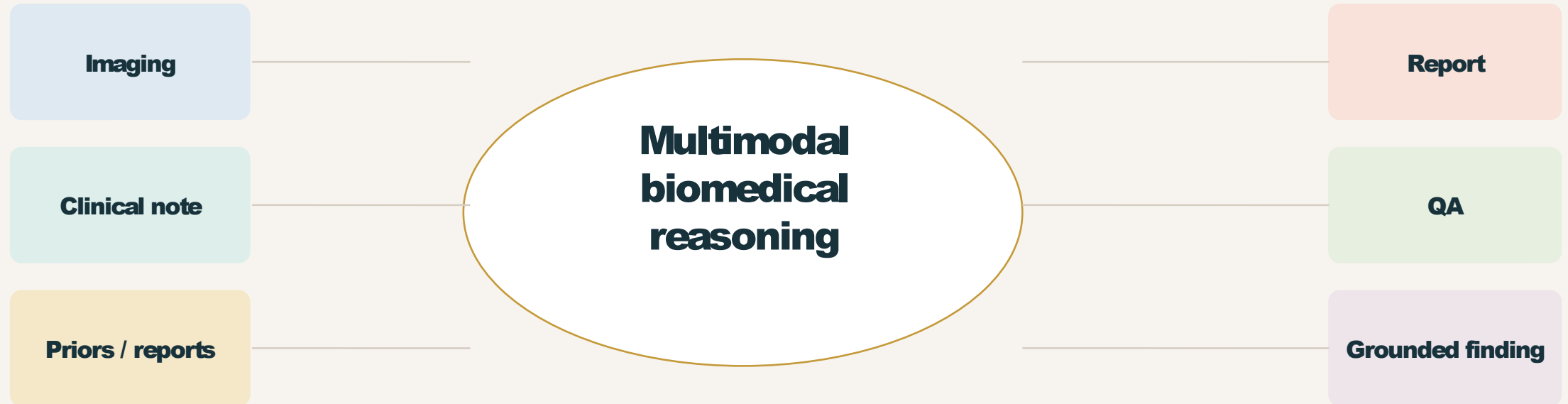
Open-source medical pretraining.

Figure: Med-PaLM benchmark and human-evaluation overview

Source citation: Singhal et al. 2023, Med-PaLM Figure 1.



Why Multimodality Matters in Medicine



Medicine is already multimodal in the workflow itself.

Common Architecture Patterns for Biomedical MLLMs



Encoders

- Image encoder
- Slide encoder
- Sequence or molecular encoder
- Clinical text encoder

Connector

- Projector
- Q-former
- Cross-attention bridge
- Token adapter

LLM core

- Instruction tuning
- Multimodal alignment
- Grounded prompts
- Context integration

Outputs

- Answer
- Report
- Localization cue
- Reasoning scaffold
- Workflow action

Key design question: how much of the performance comes from scale versus better grounding and better connectors?

Early Medical MLLMs

LLaVA-Med and Med-Flamingo

LLaVA-Med

Data focus

Biomedical figures and images

Training recipe

Rapid biomedical adaptation of a vision-language assistant

Teaching strength

Simple, memorable bridge from general VLMs into biomedicine

Caution

A medical VQA score is not a deployment argument

Med-Flamingo

Data focus

Multimodal medical data with clinician-facing evaluation

Training recipe

Few-shot multimodal learner

Teaching strength

Good for explaining few-shot transfer in medicine

Caution

Clinical utility still depends on task scope and workflow

Generalist Biomedical Multimodal Models

Med-PaLM M and BiomedGPT

[Figure — see notes]

Figure: Med-PaLM M modality-task map

Source citation: Tu et al. 2024, Med-PaLM M Figure 1.

Breadth

One model family spans many biomedical tasks and modalities.

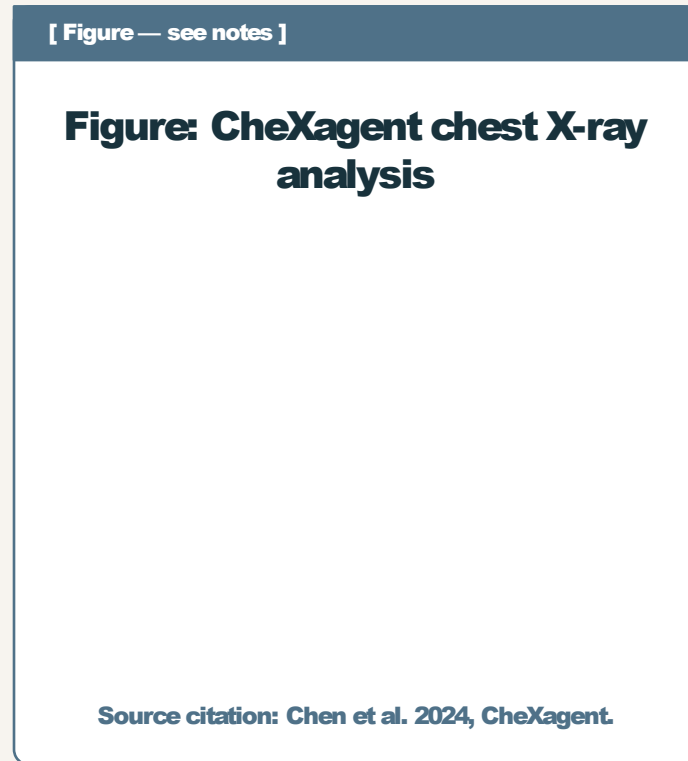
Depth

A broad system still may underperform a tightly scoped domain model on a narrow workflow.

Lecture use

Focused Chest X-Ray Assistants

CheXagent



**Focused radiology
foundation model**

Findings

Question answering

Report support

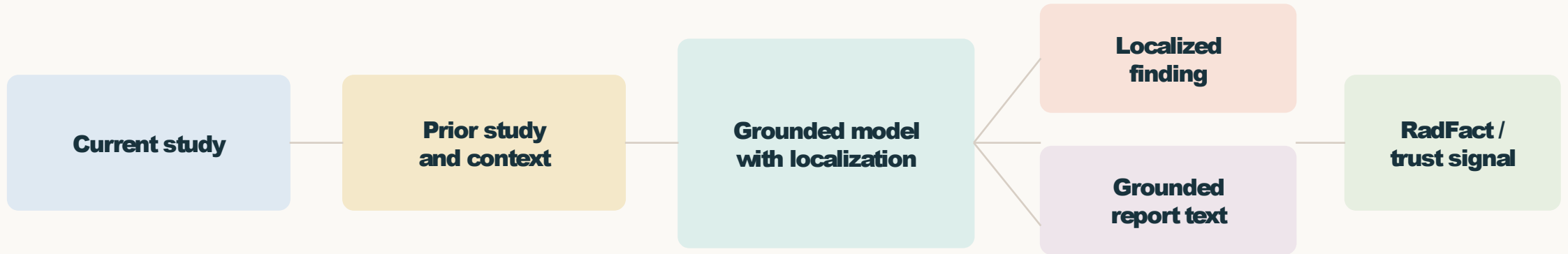
Teaching point

A narrower system can win on clarity of scope, evaluation, and workflow fit.

Generalist is not automatically better.

Grounded Radiology Report Generation

MAIRA-2



Why this slide matters

MAIRA-2 is a strong teaching example because it moves from generic generation toward grounded, inspectable report support.

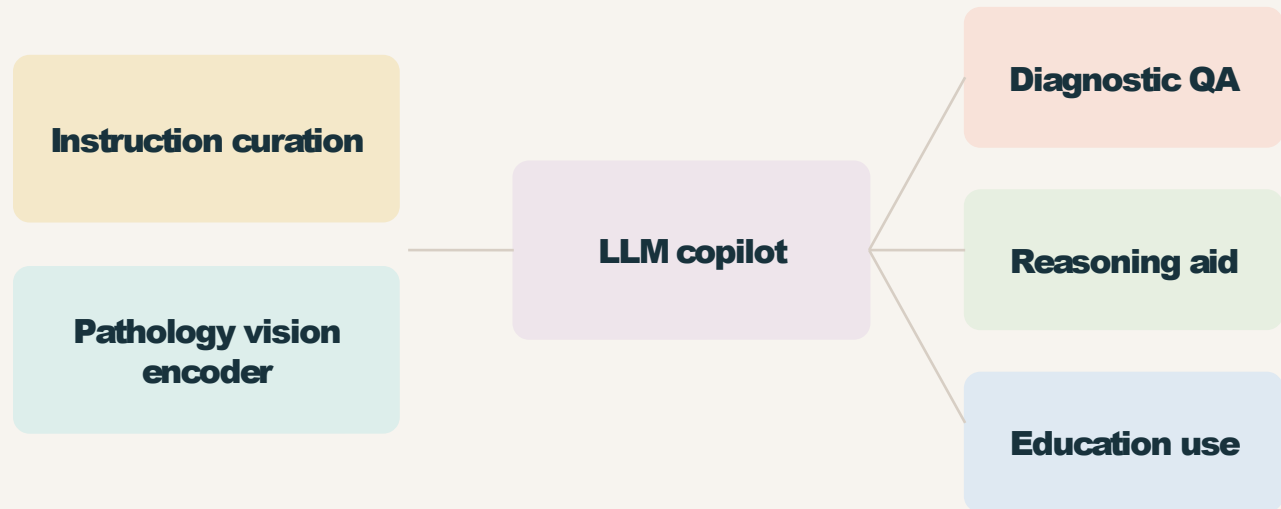
Pathology Copilots

PathChat

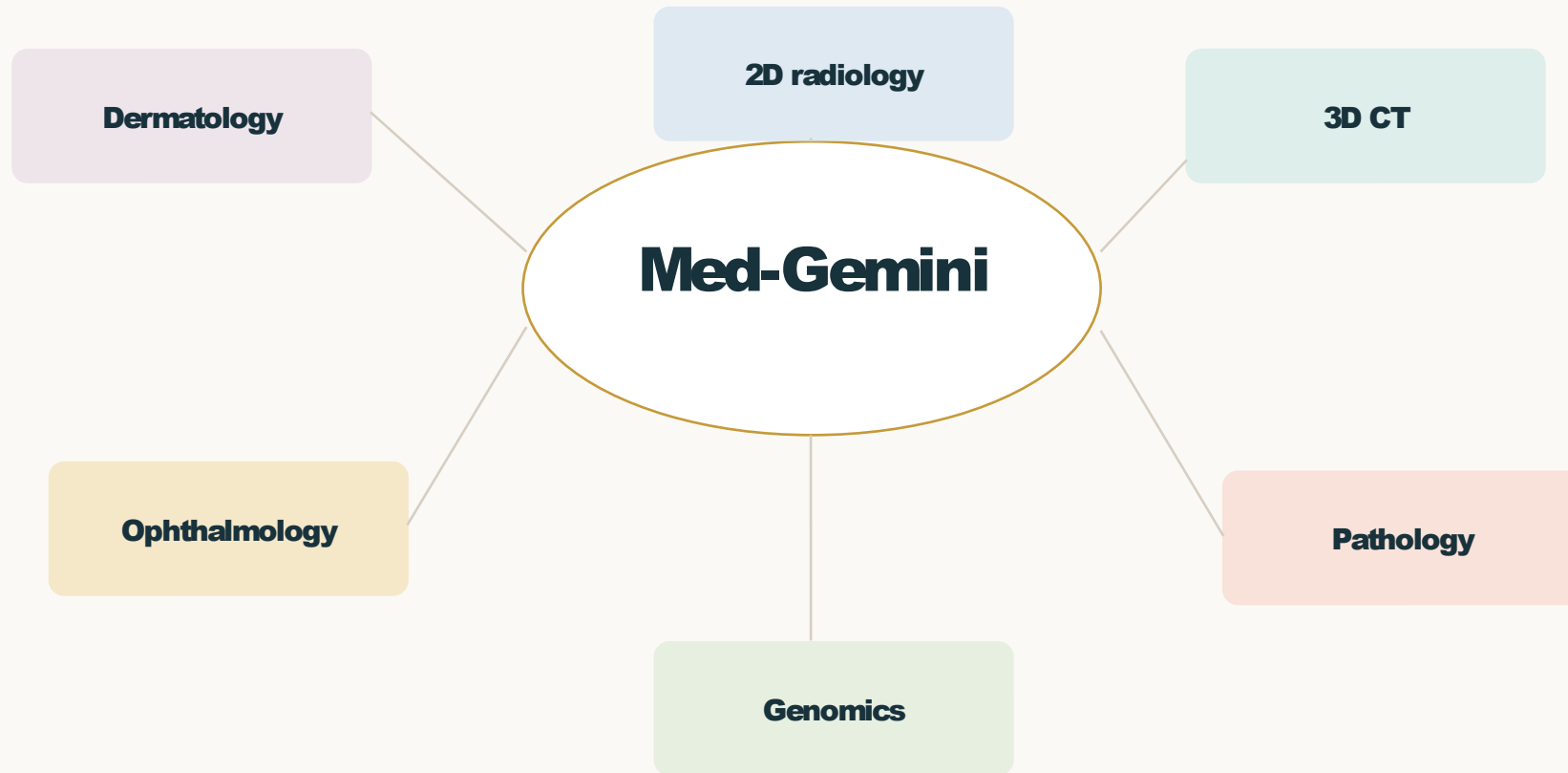
[Figure — see notes]

Figure: PathChat architecture overview

Source citation: Ding et al. 2024, PathChat Figure 1.



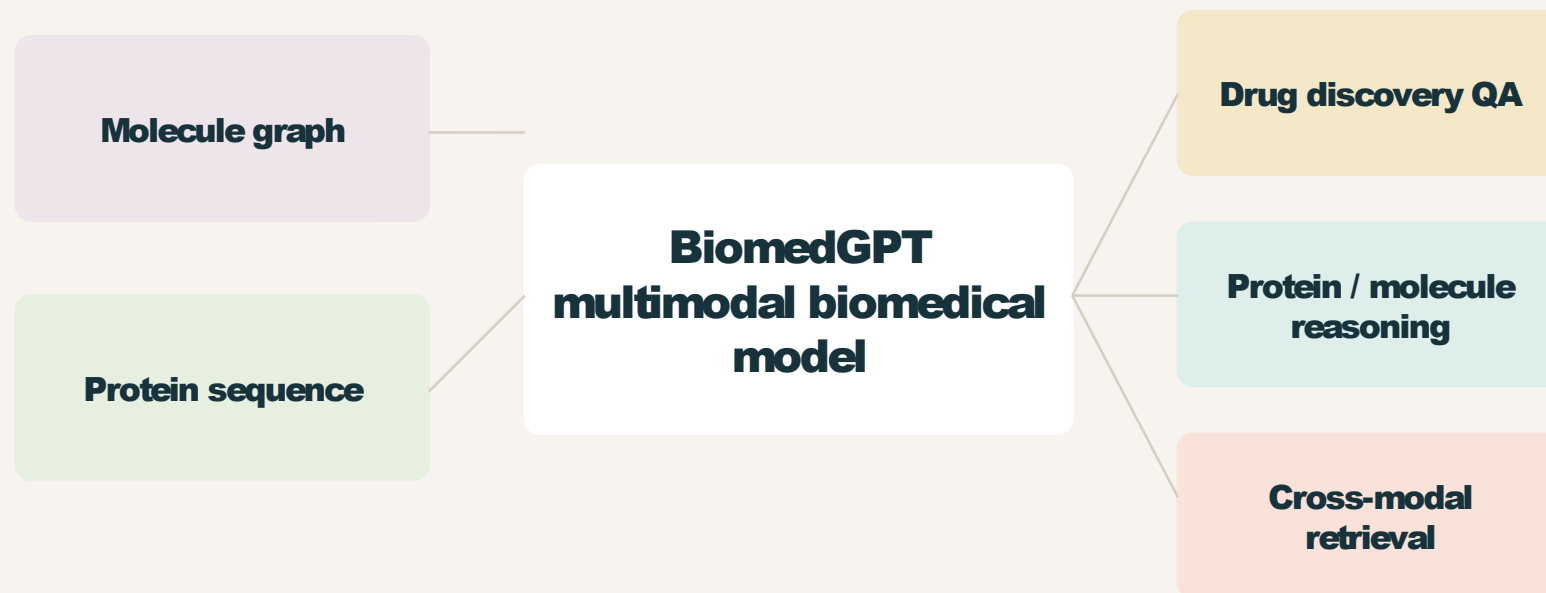
Frontier Multimodal Scaling



Question to raise in class: does “generalist medical AI” mean one architecture, one benchmark suite, or one deployment story?

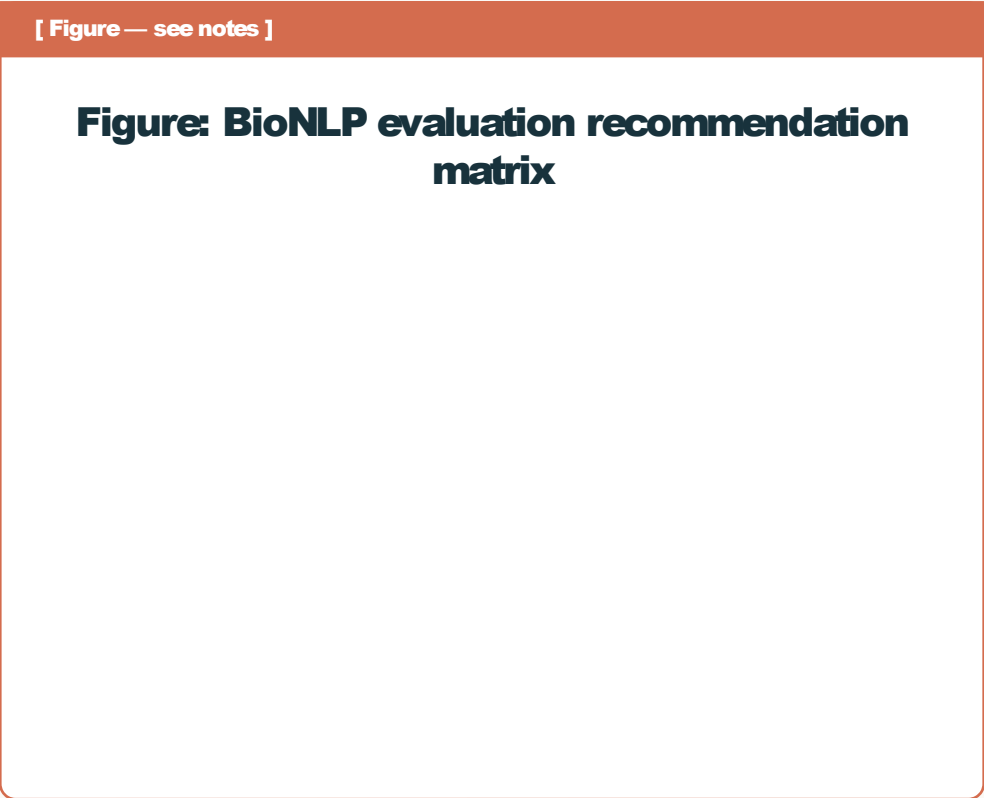
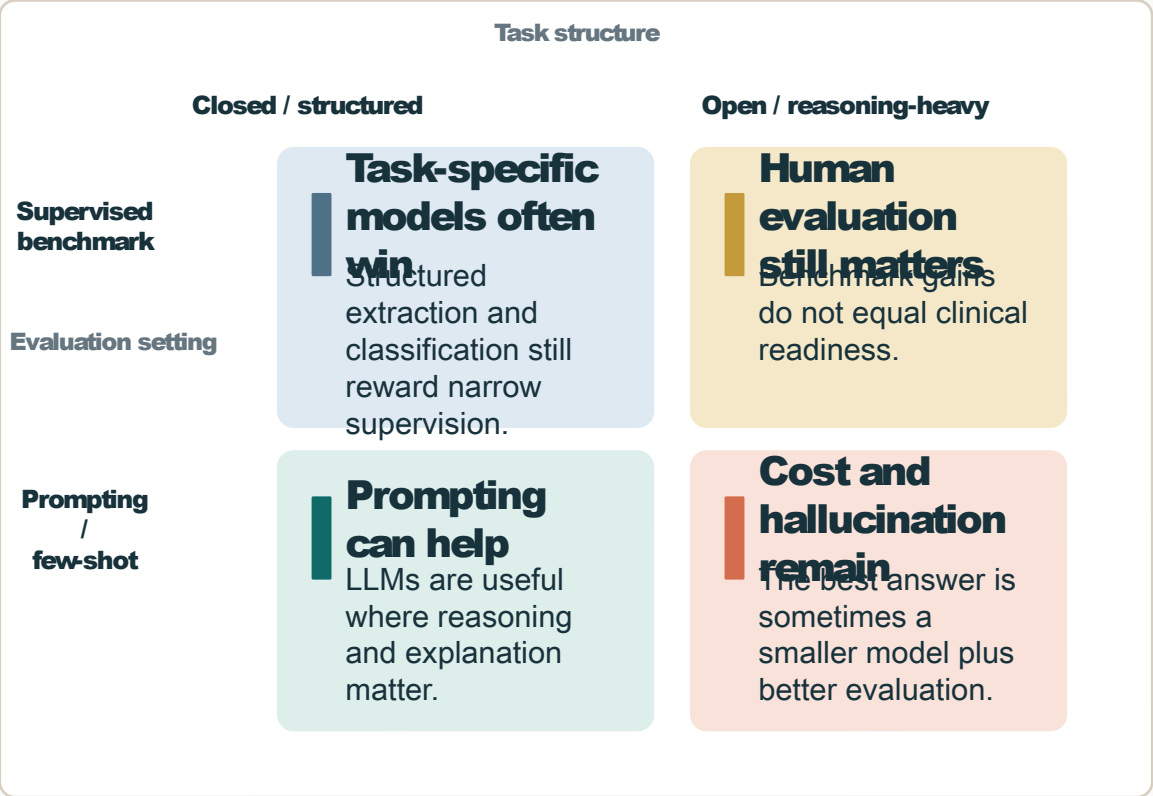
Beyond Clinical Images

Molecules, proteins, and language



Course bridge

Evaluation Reality Check



Safety, Grounding, Privacy, and Workflow Fit



Factuality

Does the answer stay tied to evidence rather than fluent pattern completion?

Traceability

Can a clinician inspect where the conclusion came from?

Privacy

Are the training data, prompts, and outputs appropriate for clinical settings?

Workflow fit

Would a real user trust and adopt this in the clinical workflow?

In-Class Discussion or Short Activity



Prompt A

Which is more deployable today:

A focused chest X-ray assistant
or
A broad multimodal medical copilot?

What would you need to evaluate before saying yes?

Prompt B

If you were reviewing a pathology copilot for education rather than diagnosis, what additional metrics would you ask for beyond benchmark accuracy?

Prompt C

Name one biomedical task where a smaller task-specific model is still the better engineering choice.

Why?

Take-Home Messages

1. Domain matters

Biomedical LLMs started by specializing language models to a domain with different vocabulary, style, and tasks.

2. Generation changed the frame

BioGPT, Med-PaLM, and related systems pushed biomedical NLP toward assistant-style generation and reasoning.

3. Medicine is multimodal

Radiology, pathology, genomics, and discovery workflows make multimodal modeling a natural next step.

4. Deployment is the hard part

Grounding, evaluation, privacy, and workflow fit are the real separators between demos and usable systems.